

The Impact of Organizational Culture on Agile Method Use

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Abstract

Agile method proponents believe that organizational culture has an effect on the extent to which an agile method is used. Research into the relationship between organizational culture and information systems development methodology deployment has been explored by others using the Competing Values Framework (CVF). However this relationship has not been explored with respect to the agile development methodologies. Based on a multi-case study of nine projects we show that specific organizational culture factors correlate with effective use of an agile method. Our results contribute to the literature on organizational culture and system development methodology use.

1. Introduction

Agile methods are a group of system development methodologies that share a common philosophy, values, and goals. The primary goal of all agile methods is to deliver software products quickly, and to adapt to changes in the process, product, environment, or other project contingencies [1]. Agile methods accommodate this change by employing a rapid iterative and incremental development process with high levels of communication and customer involvement.

While evidence suggests that agile methods have been adopted in a wide variety of organizational settings [4, 5], such methods are assumed to be more suited to certain organizational environments than others. In order to better understand the relationship between agile methods and organizational setting, we conducted a multi-case study of software development projects. Our study focused specifically on organizational culture, and the relationships between different aspects of organizational culture and the use of agile methods. The research question being investigated in this study was: *to what extent is the use*

of an agile method related to the culture of the organization?

Organizational culture is a major research area in organization studies [2] and is considered important in successful agile method use [3-8]. We used the Competing Values Framework (CVF) instrument [9] to assess the perception of project leaders as to the organizational culture of their project workplace. We also developed a technique for determining the extent to which agile methods were being used in a particular project. Using non-parametric statistical analysis of the data, we examined the correlations between specific organizational culture factors and the extent of agile method use.

The remainder of this paper is organized as follows: first we introduce the agile methods followed by a brief review of the literature on studies of system development methodologies and organizational culture. Then we describe the research problem and the case study method used to address the problem. We follow this with our data analysis method, results and a discussion of the results. We address the study limitations; provide a conclusion and indications for future research in this area.

2. Agile Methods

Initially agile methods were called lightweight methods to distinguish them from traditional “heavyweight” methods; method “heaviness” is a characteristic of prescriptive approaches requiring the production of many non-software artifacts, mainly documentation, during development. Examples of heavyweight methodologies are SSADM [10], Information Engineering [11], Unified Software Development Process [12], and OPEN [13]. The first agile method was published in 1995. Subsequently a number of other lightweight methods were published, all with a focus on expediting software development in the business and technology environment of the late 1990s and early 2000s. (Note that we use the terms

software development method and system development methodology interchangeably in this paper).

A manifesto for agile software development was published in 2001, listing the principles common to all agile methods. The manifesto authors were individuals who had previously proposed and published individual lightweight methods with similar characteristics. Each method was based on practitioner experience and evolutionary development practices with a focus on early delivery of quality software [14]. When the manifesto was written a label (agile) from the field of flexible manufacturing was adopted, and the lightweight methods became agile methods.

The techniques used within each agile method, however, are not new; it is the philosophy and the combination of techniques which is unique. Abrahamsson et al. [15] investigated the evolution of agile methods. The four main influences on agile software development were; object-orientation, evolutionary development, internet technologies, and methodology engineering. The object-oriented influences were from UML [16] and the Rational Unified Process [17]. Evolutionary approaches include the influence of Boehm's spiral model [18], prototyping and RAD. The internet technologies include ideas from Microsoft's sync and stabilise method [19] and Internet speed development [20-22]. The methodology engineering ideas came from Kumar and Welke [23] and amethodical development [24-26]. This shows that agile methods are an amalgamation of previous methodologies, ideas and practices from the software engineering field.

Influences from information systems on the development of agile methods include the Participatory Design movement [27], and ideas from soft systems methodology [28]. Participatory Design [27] is a Scandinavian movement focusing on stakeholder participation in the development of information technology solutions. The influence of Checkland's soft systems methodology [28] and ideas of human activity systems being holistic and evincing emergent properties appear in the Crystal methods [3] and Adaptive Systems Development [29].

Like any system development methodology each agile method consists of a coherent philosophy and set of techniques, development phases, and guidelines for appropriate use. Each agile method however is distinct from the others; each has a different focus. For example Extreme Programming (XP) is designed specifically for high change environments, satisfying customer needs and maintaining effective small teams [7]. Scrum focuses on project management of iterative development [30]; Dynamic Systems Development Method (DSDM) is a framework for rapid application development; Crystal methods offer techniques for

designing a methodology to suit a specific project [3]; and Adaptive System Development (ASD) is a framework for managing software projects that are under intense time pressure and have rapidly changing requirements [29].

This brief summary of the agile methods shows that they embody existing knowledge about system development as formulated in the fields of software engineering and information systems development but they extend this to accommodate a faster development pace under conditions of high change. These methods have generated a large body of research but as Dyba and Dingosyr [32] found after an extensive review of agile method literature, only 33 rigorous empirical studies have been published and 75% of the studies they identified investigated XP. They also found that research into these methods is nascent [31] and requires further empirical studies using rigorous research methods. This study aims to contribute to the agile methods research stream by focusing on the relationship between organizational culture and agile method use. We first identify initial culture concepts and then assess their impact in industry-based software development projects.

3. Organizational Culture and Information Systems Development

Organizational culture is a shared belief system that permeates an organization or subunit and ultimately influences the actions of people and work groups. The literature provides many definitions (for example Hutch [2] reports six and Leidner and Kayworth [33] report 164). Schein's [34] three level conceptualization of organizational culture forms the basis of many recent studies. He defined three levels of culture: artifacts and creations, values, and basic assumptions, and determined that values are the most readily testable. Cameron and Quinn [9] developed their competing values framework (CVF) based on organizational values. The questionnaire associated with their framework provides 24 validated quantitative measures of perceived culture values. We used a slightly modified version of this questionnaire in our study.

Organizational culture and information systems development was identified as a research theme by Leidner and Kayworth [33] who reviewed empirical studies of organizational culture in information systems research at national and organizational levels. Such studies were found to be "concerned with the influence of values on the process of systems development" [33, p. 365]. An example of this is shown in the study by Iivari and Huisman [35] who

investigated the relationship between organizational culture and information systems methodology deployment. They investigated the deployment of classical structured methodologies and found that where organizational culture was more hierarchical the developers reported higher deployment. No such positive relationship was found for IT managers. They omitted agile methods in their study, and noted that this was an area of future interest.

Agile method authors clearly state that the organizational culture in which the agile method is embedded could have an impact on its use. They also make it clear that certain types of culture would be detrimental or make it impossible to successfully use an agile method. For example, Beck states “If an organization’s actual values are secrecy, isolation, complexity, timidity, and disrespect; suddenly expressing the opposite values through a set of new practices will cause trouble rather than create improvement” [36, p. 144].

To investigate this aspect of system development we developed a set of organizational culture factors that are considered necessary for appropriate use of an agile method (see table 1). This set was identified from a review of literature on the organizational culture for agile methods and a detailed analysis of five of the earliest published agile methods (XP [7], Scrum [30], DSDM [37], ASD [38], and Crystal Methods [3]).

Table 1. Initial model of the organizational culture for agile methods – from the literature

	Organizational culture factor	Source
1	The organization values feedback and learning.	[3, 7, 30, 37, 38]
2	Social interaction in the organization is trustful, collaborative, and competent.	[6]
3	The organization values teamwork.	[3, 7, 30, 37, 38]
4	The organization is flexible and participative and encourages social interaction.	[4]
5	The project manager acts as a facilitator.	[4]
6	The organization enables empowerment of people.	[3, 7, 30, 37-39]
7	The management style is that of leadership and collaboration.	[4]
8	The organization values face-to-face communication.	[3, 7, 30, 37, 38]
9	Communication in the organization is informal.	[4, 40]

4. Research Method

We addressed the research question using a multi-case study of nine software development projects.

Within this study we investigated the target environment factors for agile methods by assessing both the extent of method tailoring on projects, and project environment factors, including organizational culture. The case study approach was considered appropriate for this research for a number of reasons. Case study research is considered particularly appropriate for the study of information systems development [41] and it is also appropriate in software engineering research [42]. Yin argues that case studies are the “preferred strategy when ‘how’ or ‘why’ questions are posed” [43, p. 1], as is the case here. Real-life context, contemporary events and a lack of control of variables by the researcher are also situations where case study research is appropriate [43]. This study of systems development projects and the environment of agile development methods met these criteria.

4.1. Unit of analysis

The unit of analysis in case study research defines the boundaries of the case investigated [43]. In this study an individual system development project comprised the unit of analysis. Each project met the following criteria:

1. The project had a well-defined purpose.
2. The project was carried out using a set of activities that aimed to produce a software application.
3. The project had both a beginning and an end date (estimated if the project was not complete at the time of the study).
4. The project was underway in that at least 1/3 of the estimated project duration was complete.
5. At least one person was working on the project.
6. The project was carried out using either:
 - An agile method.
 - A systems development methodology that was not an agile method (i.e. Rational Unified Process, another named methodology, or a documented in-house methodology).
 - No systems development methodology (i.e. ad hoc development).

4.2. Validity

Validity in case study research is concerned with construct validity, internal validity, external validity and reliability [43]. We address construct validity by using a validated questionnaire for evaluating organizational culture published by Cameron and Quinn [9]. Internal validity is considered less relevant in case study research (it is appropriate in experimental

research) [43]. Generalizability, or external validity, is a concern with case study research. There is no attempt in this study to generalize the results to a wider population, and appropriate non-parametric statistics are used. The selection of cases followed both a replication and theoretical logic allowing for generalization to theory rather than populations [43]. Reliability is concerned with ensuring that researcher bias is eliminated and other researchers would draw the same conclusions if they carried out the study. This is addressed by using a case study protocol for case selection, a questionnaire that was the same for all respondents and statistical tests to carry out a cross-case analysis. All procedures were documented.

4.3. Case design and selection

Cases were selected using a convenience sampling strategy [44]. Participants were identified by an Internet search for organizations carrying out software development in New Zealand who advertise or state, typically in practitioner journals, that they use agile methods, and through personal contacts made at developer conferences, at local IT seminars, and at a specialist developers group. Participants were selected because they met the criteria of working on a project (see unit of analysis above) and had first hand experience with the project and an overview of the process used during the whole length of the project. While we have referred to these individuals as project leaders, their roles in the workplace included business and project managers, project directors, team leaders, analyst programmers, analysts and software developers and coaches.

In case study research it is appropriate to select cases based on replication logic and theoretical logic [43]. In order to follow replication logic we selected five cases that reported using agile methods. One of these was found to be a mixed method project (agile and RUP). Theoretical replication was achieved by selecting four cases where agile methods were not used. See table 3 for details.

4.4. Preparation for data gathering

For each case organization, a questionnaire was initially administered, and was followed by a face-to-face interview. Pre-testing of questionnaires, and the use of pilot case studies are recommended for case study research [43, 44]. Cognitive pre-testing was used to ensure the questionnaire was easy to understand and did not contain any inappropriate language.

A pilot case study was used to ensure the questionnaire and the interview were carried out

effectively and efficiently and provided appropriate data. Based on this experience we decided to deliver and receive the questionnaire by post.

4.5. Data collection

The questionnaire was sent to one project leader working on each project. The questionnaire assessed the organizational culture, the nature of the agile method(s) used, and the ways in which they were used within the project.

In order to determine which agile method techniques were used during the project, the questionnaire contained a list of all techniques and major practices used in the five agile methods listed above. Table 2 shows the techniques assessed and how they were presented in the questionnaire. The techniques were grouped into similar types for respondents. They are not shown as sorted by the agile method to which they belong. Respondents selected the extent to which each technique was used on a four point Likert scale.

Table 2. Agile method techniques assessed

Agile method techniques and practices	
1	Concurrent development (analysis, design, code, and test carried out simultaneously)
2	Iterative development
3	Time boxing (iterations of set length)
4	Incremental development (small releases)
5	Evolutionary prototyping
6	Small releases of software product
7	Component development (sets of features are developed concurrently)
8	Test first development
9	Daily builds of complete system
10	Automated regression testing
11	Refactoring of code
12	Testing throughout each iteration (including unit, integration and acceptance testing)
13	Software inspections
14	Customer on-site
15	Method coach on site
16	Tester(s) colocated with team
17	Customer focus groups
18	Rooms organised for pair programming
19	Whole team works in same office or floor
20	Dedicated meeting space
21	Pair programming
22	Coding to an agreed standard
23	Collective ownership of code
24	40 hour week
25	Sprint Goal
26	Daily team meetings
27	Iteration planning meeting
28	Planning game (meeting with stakeholders at start of each iteration to plan scope)
29	Reflective workshops for methodology adaptation at each iteration
30	User stories

Agile method techniques and practices	
31	System metaphor developed
32	Only what has direct business value is developed
33	Requirements are prioritised with the customer or user
34	Changes to requirements are negotiated with the users to maintain time frames
35	Joint Application Development (JAD) sessions (requirements gathering sessions with selected stakeholders)
36	Design is kept as simple as possible
37	Coded solution is kept as simple as possible
38	Risk assessment at each iteration
39	Product Backlog
40	Sprint Backlog
41	Release Backlog
42	Milestones to track progress
43	Product Backlog Graph metric
44	Sprint Backlog Graph metric
45	Function point counts

4.6. Data analysis

The nine projects, each of which met the unit of analysis criteria, addressed a variety of business activities within both small and medium sized enterprises from both the government and private sectors. Eight projects were carried out in New Zealand and one in the United Kingdom. Details are provided in table 3.

Table 3. Project and organization detail

Project	Method	OT	Size	Business activity
Alpha	XP	P	100-150	Control system
Beta	XP	SOE	150-200	Information presentation
Delta	XP/Scrum	G	13000-14000	Web development
Zeta	DSDM	P	300	MIS
Theta	RUP/XP	P	1400-1500	MIS
Iota	RUP	G	600	Transaction processing
Rho	Ad hoc	G	4500-5000	MIS
Tau	Ad hoc	P	5-10	Shrink wrapped software
Chi	Ad hoc	G	300-400	MIS

Key
 OT – Organization type
 Size – Rounded estimates of total number of employees in organisation
 G – Government department or fully government funded organization
 SOE – State Funded Organization
 P – Privately owned company
 DSS – Decision Support System
 MIS – Management Information System

The percentage of agile method usage for each projects was calculated as follows:

$$\% \text{ Agile method usage} = (\sum Tq / 3Tqm) \times 100$$

where

Tq = quantity of techniques used

Tqm = quantity of techniques in the method

Quantity was 0, 1, 2, or 3 (never used, seldom used, often used, always used the technique) as assessed by the project leader in the questionnaire. For agile method projects only those techniques that belong to the selected method were included (i.e. if the respondent selected techniques that do not belong to their nominated method then this technique was not included in the summation).

To compare all of the projects, both agile and non-agile, it was necessary to calculate a method usage value for each of the non-agile projects. We did this by calculating, for each non-agile project, their usage of each of the agile methods techniques. This provided five sets of data, each set comparing a non-agile project with the extent of method usage of each of the five methods; the XP method, the DSDM method, the Scrum method, the Crystal method, and the ASD method. Table 4 shows the extent to which each of the non-agile projects used each of the methods. The sum of the ranks shows that XP is the highest ranked method (i.e. more XP techniques were used on the non-agile projects than techniques from the other agile methods). So for the non-agile projects XP was nominated as the method for the project. The XP usage value was used for subsequent comparison of the projects. This meant that a comparison of the method usage and the organizational factors in the initial model could be carried out for all projects both agile and non-agile.

Table 4. Agile method usage for non-agile projects

	Project				
	Iota	Tau	Chi	Rho	$\sum$ ranks
	Method used on project				
Usage (%)	RUP	Ad hoc	Ad hoc	Ad hoc	
XP	40 (3)	54 (1)	30 (2)	18 (2)	8
DSDM	38 (4)	54 (1)	29 (3)	19 (1)	9
Scrum	36 (5)	38 (5)	21 (5)	3 (3)	18
Crystal	50 (1)	43 (4)	23 (4)	0 (4)	13
ASD	42 (2)	50 (3)	31 (1)	0 (4)	10

Note: Bracketed number is the rank from lowest usage (5) to highest (1)

An agile method usage value for each of the nine projects as shown in table 4 and table 5.

Table 5. Agile method usage for agile projects

	Project				
	Alpha	Zeta	Delta	Theta	Beta
	Method used on project				
Usage (%)	XP	DSDM	XP Scrum	RUP XP	XP
XP	96	86	67	63	56
DSDM	52	94	65	79	65
Scrum	51	77	67	49	41
Crystal	70	80	57	73	60
ASD	56	83	58	78	53

Each of the organizational culture factors shown in table 1 above was investigated. We used the adapted CVM instrument to address items 1, 2, 4, 5, 6 and 8. We used our own questions to address items 3, 7, 8 and 9. More than one question on the questionnaire was used to assess each of the factors found in the literature. All cases were then compared using non-parametric statistics to investigate the correlation between the organizational culture factors and usage of the agile method. The statistics used were Spearman's correlation coefficient (for ordinal data), and Pearson's rank correlation coefficient (for ratio data).

5. Results

We found a number of organizational culture factors that show statistically significant correlations with agile method usage (see table 6).

Table 6. Summary of results for all CVM items

CVF Item	CVF Question wording - abbreviated	Signifi - cance level [#]
A*	Organisation is personal place...	0.05
B*	Organisation is dynamic entrepreneurial..	0.05
C**	Organisation is results oriented...	0.05
D	Organisation is controlled...	
E*	Leadership is mentoring...	0.01
F**	Leadership is entrepreneurial...	0.01
G	Leadership is no-nonsense...	
H	Leadership is coordinated...	
I*	Teamwork...	0.05
J	Individual risk-taking...	
K	Hard-driving competitiveness...	
L	Security of employment...	
M**	Glue is loyalty...	0.05
N	Glue is innovation...	
O	Glue is achievement...	
P	Glues is rules...	
Q*	Emphasis on human development...	0.01
R	Emphasis on new resources...	
S	Emphasis on competitive actions...	
T	Emphasis on permanence...	
U*	Success is human resource development..	
V	Success is newest products...	

CVF Item	CVF Question wording - abbreviated	Signifi - cance level [#]
W	Success is winning...	
X	Success is efficiency...	
Key [#] Using Spearman's rank correlation coefficient on all projects (N=9, 2-tailed test) * Also identified in the literature on agile methods. ** Additional factor not present in the initial model (table 1)		

Table 7 shows the mapping of initial model factors and how they were assessed using CVF items.

Table 7. Mapping of initial model factors to CVF factors

	Organizational culture factor	CVF item
1	The organization values feedback and learning.	E, Q
2	Social interaction in the organization is trustful, collaborative, and competent.	Q
3	The organization values teamwork.	I, U
4	The organization is flexible and participative and encourages social interaction.	I
5	The project manager acts as a facilitator.	E
6	The organization enables empowerment of people.	B, Own*
7	The management style is that of leadership and collaboration.	A, E
8	The organization values face-to-face communication.	Own**
9	Communication in the organization is informal.	Own**
Key *We used the question: "Are team members actively involved in making decisions, formally or informally, about how to deal with problems arising within the project? (Scale: almost never, seldom, sometimes, often, always). ** All projects reported either balanced (between formal and informal communication) or mainly informal or informal communication		

We consolidated items that were assessed using the same CVF factors where those factors were statistically significant. Thus items 1, 2, 5, 7 in table 7 were amalgamated. Also, items 3 and 4 were amalgamated, and 8 and 9 were removed. Items 8 and 9 we found were reported on all projects, agile and non-agile.

We found three additional factors in the CVF instrument that gave statistically significant correlation with agile method usage (see table 6). The factors were:

1. The organization is results oriented,
2. The leadership in this organization is entrepreneurial, innovative, and risk taking, and
3. The organization is based on loyalty and mutual trust and commitment.

The end result of this analysis is shown in table 8.

Table 8. Organizational culture factors that correlate significantly with agile method usage

	Organizational culture factors
1	The organization values feedback and learning. Social interaction in the organization is trustful, collaborative, and competent. The project manager acts as a facilitator. The management style is that of leadership and collaboration.
2	The organization values teamwork is flexible and participative and encourages social interaction.
3	The organization enables empowerment of people.
4	The organization is results oriented.
5	The leadership in the organization is entrepreneurial, innovative, and risk taking.
6	The organization is based on loyalty and mutual trust and commitment.

6. Discussion

This empirical study provides evidence regarding the relationship between organizational culture and use of agile method techniques. We found a statistically significant correlation between the following organizational culture factors and agile method use: the organization values feedback and learning; social interaction in the organization is trustful, collaborative, and competent; the project manager acts as a facilitator; the management style is that of leadership and collaboration; the organization values teamwork is flexible and participative and encourages social interaction; the organization enables empowerment of people; the organization is results oriented; leadership in the organization is entrepreneurial, innovative, and risk taking; and the organization is based on loyalty and mutual trust and commitment.

It is important to note that on both agile and non-agile projects, respondents reported that they use informal, face-to-face communication. This may be due to the size of the project, or the national culture, or could be due to some other factors. These factors may affect agile method usage but they are not unique to agile projects.

Our results stand in contrast to two previous studies of organizational culture and systems development

methodology use. Chow and Cao [45] investigated organizational culture (specifically a cooperative, oral culture where managers are adaptive and teamwork is coherent and self organizing) as possible factors impacting on the success of agile software projects. They had 109 respondents, one per agile software project, and found no statistical support for the effect of their nominated organizational culture factors on the perceived success of the projects. Our research does not concur with their findings. Iivari and Huisman [35] investigated the affect of hierarchical and rational cultures on systems development methodology deployment. Their study included developers and IT managers, and they were concerned about traditional methodologies, not agile methods. They found a positive relationship between the hierarchical cultural orientation and traditional methodology deployment within the software developer groups. Although our study is small, our findings together with those of Iivari and Huisman suggest that traditional systems development methodologies may be more accepted in hierarchical rational organizations and agile methods more acceptable in low formality organizations. Further research is needed to investigate this potentially important relationship.

7. Limitations

Correlation does not imply causality so we cannot state either that the presence of the above organizational culture factors has an effect on agile method usage or that the high use of agile method techniques has an effect on the organizational culture. Neither can we generalize these results to all software development projects. Further studies are needed to verify our results. However, these results do provide additional support for the argument that organizational culture is a factor in agile method use.

We limited our examination to the five earliest published agile methods. If we had selected different agile methods, later versions or a greater number of agile methods the study may have provided different answers.

Furthermore, we used a single data source (respondent) in each project. A broader sample of respondents, especially team members, would have strengthened the study by providing stronger evidence regarding individual perceptions concerning organizational culture.

8. Conclusions

Organizational culture is considered to be a factor effecting successful adoption of an agile method. We investigated the relationship between organizational

culture and agile method usage within a multi-case study of nine software development projects. Cross-case analysis using non-parametric correlation statistics was employed to explore the relationships. A specific set of organizational culture factors were found to be related to the use of the agile methods we studied. The greater the extent or presence of these factors in the organizations studied, the higher was their agile method usage value (i.e., the extent of use of specific agile method techniques used on the project).

Care must be used in interpreting the results. While our tentative presumption is that specific organizational culture traits lead to the adoption of certain agile method techniques, it may in fact work the other way around: adoption of agile method techniques actually produces changes in organizational culture. Longitudinal studies may be needed to fully understand the directionality of these relationships.

The information provided by this study is important for software developers including those planning to adopt an agile method and those trying to understand the limits on their likely usage of an agile method caused by an adverse organizational culture. Although the results are not generalizable to all projects using an agile method, the influence of the organizational culture factors should be seriously considered.

Organizational culture and the deployment and effective use of systems development methodologies are related. Further rigorous empirical studies of this relationship that investigate individual system development methodologies and different development approaches [46] would be valuable for both researchers and practitioners with an interest in system development. Current studies focus on the impact of the organizational culture on the methodology but does the adoption of a system development methodology affect the organizational culture of a system development organization? We see this as a research direction informing the field of information systems development.

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